

AI Infrastructure Investment

Analysis of IT Asset Useful Life Disclosures

S&P; 1500 — SEC 10-K Filings, 2013–2024

Sample	S&P 1500 — 1,687 unique firms
Observations	18,336 firm-years (2013–2024)
Extraction status	OK: 17,415 Errors: 921
Useful life disclosed	3,879 firm-years (21.2%)
AI infra mentions	1,796 firm-years (9.8%)
Algo policy changes	130 detected (2013–2024)
Report date	June 01, 2026

1. Useful Life Disclosure Rates (2013–2024)

The share of S&P; 1500 firms that explicitly disclose IT asset useful life durations in their 10-K filings more than doubled over the study period, rising from 14.2% in 2013 to 26.4% in 2024. The most striking feature of this time series is a structural break in 2019: disclosure rates jumped from 13.3% in 2018 to 22.1% in 2019 — an increase of nearly 9 percentage points in a single year.

This 2019 break coincides with the mandatory adoption of ASC 842 (Lease Accounting), which required companies to conduct a comprehensive review of their capitalized asset policies. Many firms that had previously disclosed useful life estimates only informally were obligated to formalize them, producing a lasting shift in disclosure practice that persisted through 2024.

Year	Firms with disclosure	Total firms	Disclosure rate	Change vs prior year
2013	180	1264	14.2%	—
2014	178	1306	13.6%	-0.6pp ↓
2015	171	1345	12.7%	-0.9pp ↓
2016	172	1381	12.5%	-0.3pp
2017	169	1427	11.8%	-0.6pp ↓
2018	195	1465	13.3%	+1.5pp ↑
2019	329	1486	22.1%	+8.8pp ↑
2020	368	1505	24.5%	+2.3pp ↑
2021	401	1537	26.1%	+1.6pp ↑
2022	397	1557	25.5%	-0.6pp ↓
2023	405	1568	25.8%	+0.3pp
2024	416	1574	26.4%	+0.6pp

Note: The 2019 jump (+8.8pp) is the largest single-year increase in the dataset and likely reflects the ASC 842 adoption effect rather than genuine changes in disclosure practice.

2. IT Asset Useful Life Duration Trends

Among firms that disclose useful life estimates, the average maximum duration has increased from 12.3 years in 2013 to 13.8 years in 2023–2024. The median remains stable at 10 years throughout the period, indicating that the shift is driven by a subset of firms extending durations significantly rather than a broad market-wide change.

Year	Mean max (yrs)	Median max (yrs)	N firms
2013	12.3	10	140
2014	12.5	10	139
2015	13.3	10	132
2016	12.0	10	136
2017	12.2	10	132
2018	13.2	10	155
2019	13.2	10	253
2020	13.2	10	280
2021	12.8	10	297
2022	13.0	10	300
2023	13.8	10	315
2024	13.8	10	323

2.1 Notable Individual Policy Changes

The algorithmic comparison of year-over-year values identifies 130 significant duration changes ($\Delta \geq 1$ year) across the sample — 85 extensions and 45 reductions. The largest changes tend to be linked to major M&A; activity or asset reclassification events, not organic policy revision. Below are the most significant examples:

Firm	Year	From (yrs)	To (yrs)	Δ	Context (verbatim excerpt)
MKSI	2018	10	50	40	Depreciation is provided on the straight-line method over the estimated useful lives of te...
KLXE	2023	3	40	37	Property and equipment are stated at cost and depreciated generally under the straight-lin...
PII	2023	7	40	33	Property and equipment. Property and equipment is stated at historical cost. Depreciation ...
BDX	2024	13	45	32	Property, plant and equipment are stated at cost, less accumulated depreciation and amorti...
BEN	2019	5	35	30	Property and Equipment, net are recorded at cost and depreciated using the straight-line m...
NWSA	2024	20	50	30	Original Useful Lives As of June 30, 2024 2023 (in millions) Property, plant and equipment...

Example — Broadcom (AVGO) FY2022: extended max useful life from 10 to 40 years following the \$61bn acquisition of VMware, reducing annual depreciation charges on the acquired asset base.

Example — MKS Instruments (MKSI) FY2018: extended from 10 to 50 years in the year of the Newport Corporation integration. The extension was reversed in 2024 (50→15 years) when assets were fully amortized.

2.2 Changes by Year

Extensions are consistently more frequent than reductions, with a notable acceleration in 2019 (+14 extensions, matching the disclosure rate break) and again in 2023 (+13 extensions). The 2022 year shows the highest number of reductions (-10), coinciding with rising interest rates prompting some firms to reassess asset valuations downward.

Year	Extensions	Reductions	Net change
2013	—	—	—
2014	+5	-0	+5
2015	+5	-1	+4
2016	+6	-3	+3
2017	+4	-3	+1
2018	+8	-4	+4
2019	+14	-5	+9
2020	+7	-2	+5
2021	+6	-5	+1
2022	+9	-10	-1
2023	+13	-5	+8
2024	+8	-7	+1

3. AI Infrastructure Disclosures

The share of firms mentioning AI infrastructure in their 10-K filings grew from 7.2% in 2013 to 15.4% in 2024 — a 114% relative increase. The growth trajectory can be divided into three phases: a flat early period (2013–2018, 7–9%), a moderate acceleration coinciding with cloud buildout (2019–2022, 11–12%), and a sharp acceleration post-ChatGPT (2023–2024, 13–15%).

Year	AI mentions	Total firms	Rate	Item 8	MD&A
2013	91	1264	7.2%	29	62
2014	97	1306	7.4%	34	63
2015	94	1345	7.0%	34	60
2016	103	1381	7.5%	41	62
2017	107	1427	7.5%	43	64
2018	135	1465	9.2%	60	75
2019	173	1486	11.6%	87	86
2020	170	1505	11.3%	97	73
2021	189	1537	12.3%	104	85
2022	193	1557	12.4%	105	88
2023	202	1568	12.9%	117	85
2024	242	1574	15.4%	127	115

3.1 Shift from MD&A; to Item 8

A critical qualitative shift is visible in the source of AI-related disclosures. Prior to 2019, MD&A; (Item 7) was the dominant source of AI mentions (62 MD&A; vs 29 Item 8 in 2013), reflecting AI as a strategic narrative rather than an accounting reality. From 2020 onward, Item 8 (Financial Statements) overtook MD&A; and continues to widen the gap (127 Item 8 vs 115 MD&A; in 2024). This transition signals that AI infrastructure has moved from managerial aspiration to capitalized investment on balance sheets.

3.2 NVIDIA Mentions — A Clean Signal

NVIDIA-specific mentions provide the most reliable hardware adoption signal across the full period, as the company name does not appear in non-AI contexts. The growth is steady and accelerating:

Year	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Firms	11	10	12	11	11	12	12	14	16	17	19	26

From 11 firms in 2013 to 26 firms in 2024, NVIDIA mentions more than doubled, with the steepest increase between 2022 and 2024 (+9 firms in 2 years).

4. Illustrative Company-Level Examples

4.1 AMD (Advanced Micro Devices) — A Strategic Pivot Case

AMD provides a textbook example of how 10-K disclosures track corporate AI strategy. Over 2013–2024, the company's filings reveal two synchronized transitions: a doubling of maximum useful life and a progressive shift in AI disclosure language.

Period	Max useful life	AI disclosure language
2013–2021	6 years	GPU mentions in gaming/graphics context; no AI infrastructure language
2022	15 years	"Data Center segment: server CPUs and GPUs, DPUs, FPGAs and Adaptive SoC"
2023	15 years	"Data Center: server microprocessors (CPUs), graphics processing units (GPUs)"
2024	15 years	"Data Center: Artificial Intelligence (AI) accelerators, MI300 series, Instinct"

The 2022 transition is decisive: AMD extended useful life from 6 to 15 years exactly when it formally created a dedicated Data Center segment. The `policy_change` field in the extracted data records 'No' (no explicit mention of change in the text) — illustrating why algorithmic detection is indispensable and cannot be replaced by text search alone.

4.2 Dycom Industries (DY) — Indirect AI Beneficiary

Dycom, a telecommunications construction company, exemplifies indirect AI infrastructure exposure. The company's 2024 10-K states:

"In addition, the advent of AI data centers has created expanding opportunities as hyperscalers look to connect data centers with long-haul, private, redundant fiber networks. Finally, wireless networks are deploying additional spectrum bands and equipment to more broadly and efficiently provision higher bandwidth services."

This is classified as MD&A; source, reflecting a strategic narrative rather than a balance sheet entry. Dycom does not build AI hardware — it lays the fiber that connects data centers. This type of disclosure captures the second-order effects of AI infrastructure investment that a purely hardware-focused search would miss.

4.3 Broadcom (AVGO) — M&A-Driven; Duration Extension

Broadcom extended maximum useful life from 10 to 40 years in FY2022 — a +30 year jump. This coincides with the announcement of the VMware acquisition. The verbatim reads:

"Property, plant and equipment are stated at cost less accumulated depreciation and amortization. When assets are retired or disposed of, the assets and related accumulated depreciation and amortization are removed from our property, plant and equipment balances and the resulting gain or loss is refl..."

This extension was reversed in 2024 (40→10 years), consistent with the completion of purchase price allocation after the VMware close. It illustrates a common false positive risk: M&A; accounting can produce duration swings that look like AI investment signals but are purely structural accounting events.

4.4 Dover Corporation (DOV) — Liquid Cooling Signal

Dover's 2024 MD&A; section captures an emerging infrastructure theme — liquid cooling for high-performance computing:

"The organic revenue growth was driven by robust shipment rates of single-use biopharma components and connectors used in liquid cooling of high performance computers and in data centers..."

Dover is an industrial manufacturer, not a tech company — yet its revenue is increasingly tied to data center cooling demand. This type of disclosure would be classified as 'indirect AI infrastructure' and illustrates the breadth of the AI investment footprint across sectors.

5. Conclusions

5.1 Main Findings

#	Finding	Evidence
1	Disclosure rates doubled over 2013–2024, with a structural break in 2019	14 in 2019 (26.4%); +8.8pp jump in 2019 (ASC 842 effect)
2	IT asset useful life durations are extending, driven by a hardware model shift	Median firm life: 12.3 yrs (2013) → 13.8 yrs (2024); median stable at 10 yrs
3	AI infrastructure mentions more than doubled, accelerating post-2022	752 in 2023 (2013) → 15.4% (2024); steepest growth 2022–2024
4	AI disclosures shifted from strategic narrative (MD&A) to accounting (Item 8)	2013: 62 (Item 8) vs 62 MD&A → 2024: 127 Item 8 vs 115 MD&A
5	Duration extensions are predominantly linked to M&A, hardware, and AI investments	Large M&A acquisitions: AMZN (+40), KLXE (+37), AVGO (+30) — all acquisition years
6	NVIDIA mentions grew from 11 to 26 firms (2013–2024), accelerating post-2022	10 potential hardware signal across the full sample period

5.2 Limitations

Three important limitations constrain interpretation of these results. First, the 21.2% useful life disclosure rate means the majority of observations have no duration data — the disclosed sample may not be representative of S&P; 1500 as a whole, potentially overrepresenting capital-intensive firms with more detailed accounting policies.

Second, hardware model detection (A100, B200, etc.) produces false positives before 2020, as these strings appear in other product codes. NVIDIA brand mentions are the only reliable hardware signal across the full time series.

Third, the 'policy change' field relies on explicit textual mentions in the filing. The algorithmic comparison approach (year-over-year delta) identifies 130 changes versus 1,396 flagged by text search — suggesting significant over-flagging by keyword detection. A manual review of a random sample would be needed to establish the true false positive rate.

5.3 Research Implications

The convergence of extending useful lives, formalizing AI disclosures in financial statements, and NVIDIA mentions in non-tech firms points to a broad-based capitalization of AI infrastructure across the S&P; 1500. The 2022–2024 period marks a qualitative shift: AI infrastructure is no longer only a tech-sector phenomenon but appears across energy, industrials, and telecom firms that build, power, and connect the physical AI stack.

Future work should combine this accounting data with capital expenditure data to test whether firms extending useful lives are also increasing CapEx — which would confirm the AI investment hypothesis rather than attributing changes to accounting optimization alone.